Distribution Shift Evaluation (Cityscapes) This notebook evaluates a trained Cityscapes segmentation baseline under synthetic corruptions (blur, noise, brightness) to quantify robustness.

1. Setup

In [1]:
```
from pathlib import Path
import sys
import torch

# Notebook lives in notebooks/, so parent is the project root.
ROOT = Path.cwd().parent
sys.path.append(str(ROOT))

DATA = ROOT / "data"
OUT = ROOT / "outputs"
WEIGHTS = ROOT / "weights"
OUT.mkdir(exist_ok=True)

# Prefer GPU when available.
device = "cuda" if torch.cuda.is_available() else "cpu"
print("ROOT:", ROOT)
print("DATA:", DATA)
print("WEIGHTS:", WEIGHTS)
print("device:", device)
```

```
ROOT: c:\Users\dhruv\temporal_cityscapes
DATA: c:\Users\dhruv\temporal_cityscapes\data
WEIGHTS: c:\Users\dhruv\temporal_cityscapes\weights
device: cuda
```

2. Imports

In [2]:
```
import numpy as np
import matplotlib.pyplot as plt

from torch.utils.data import DataLoader
import torchvision.transforms.functional as TF

from src.dataset_cityscapes import CityscapesTrainId
from src.train_utils import get_device, make_deeplab, evaluate, load_checkpoint
```

Load validation dataset

In [3]:
```
SIZE_HW = (512, 1024)

val_clean = CityscapesTrainId(DATA, split="val", size_hw=SIZE_HW)

def make_loader(ds, bs=6, shuffle=False):
    # Multi-worker loader is fine for the clean dataset and speeds up eval.
    return DataLoader(
```

```
        ds, batch_size=bs, shuffle=shuffle,
        num_workers=8, pin_memory=True, persistent_workers=True,
        drop_last=False
    )

loader_val_clean = make_loader(val_clean, bs=6, shuffle=False)
len(val_clean)
```

Out[3]: 500

Define corruption wrapper

In [4]:
```
class CorruptWrapper(torch.utils.data.Dataset):
    def __init__(self, base_ds, mode: str):
        self.base = base_ds
        self.mode = mode

    def __len__(self):
        return len(self.base)

    def __getitem__(self, idx):
        x, y, meta = self.base[idx]

        # Corrupt image only; keep labels untouched.
        if self.mode == "blur":
            x = TF.gaussian_blur(x, kernel_size=7, sigma=1.35)
        elif self.mode == "noise":
            x = x + 0.14 * torch.randn_like(x)
            x = torch.clamp(x, -3.0, 3.0)
        elif self.mode == "bright":
            x = x * 1.25
            x = torch.clamp(x, -3.0, 3.0)
        else:
            raise ValueError(f"Unknown corruption mode: {self.mode}")

        meta = dict(meta)
        meta["corruption"] = self.mode
        return x, y, meta
```

Load trained model checkpoint

In [5]:
```
device = get_device()

ckpt_path = WEIGHTS / "deeplabv3_cityscapes_baseline.pt"
model = make_deeplab(num_classes=19, pretrained=True).to(device)

ckpt = load_checkpoint(ckpt_path, model, map_location=device)
trained_size_hw = ckpt.get("size_hw", (512, 1024))
print("Loaded:", ckpt_path)
print("Checkpoint keys:", list(ckpt.keys()))
print("Trained size_hw:", trained_size_hw)
```

```
Loaded: c:\Users\dhruv\temporal_cityscapes\weights\deeplabv3_cityscapes_baseline.pt
Checkpoint keys: ['model', 'size_hw']
Trained size_hw: (512, 1024)
```

In [6]:

```
from torch.utils.data import DataLoader

# Safety fallback when cells are run out of order.
if "device" not in globals():
    device = get_device() if "get_device" in globals() else ("cuda" if torch.cuda.i

if "trained_size_hw" not in globals():
    trained_size_hw = SIZE_HW if "SIZE_HW" in globals() else (512, 1024)

if "CorruptWrapper" not in globals():
    raise NameError("CorruptWrapper is not defined. Run the corruption-wrapper cell

BATCH_SIZE = 4
PIN_MEMORY = (device.type == "cuda") if hasattr(device, "type") else (str(device) =

NUM_WORKERS_CLEAN = 4
NUM_WORKERS_CORRUPT = 0

# Match evaluation resolution to training when checkpoint metadata exists.
SIZE_HW = trained_size_hw
val_clean = CityscapesTrainId(DATA, split="val", size_hw=SIZE_HW)

val_blur = CorruptWrapper(val_clean, mode="blur")
val_noise = CorruptWrapper(val_clean, mode="noise")
val_bright = CorruptWrapper(val_clean, mode="bright")

loader_val_clean = DataLoader(
    val_clean,
    batch_size=BATCH_SIZE,
    shuffle=False,
    num_workers=NUM_WORKERS_CLEAN,
    pin_memory=PIN_MEMORY,
    persistent_workers=(NUM_WORKERS_CLEAN > 0),
)
loader_val_blur = DataLoader(
    val_blur,
    batch_size=BATCH_SIZE,
    shuffle=False,
    num_workers=NUM_WORKERS_CORRUPT,
    pin_memory=PIN_MEMORY,
)
loader_val_noise = DataLoader(
    val_noise,
    batch_size=BATCH_SIZE,
    shuffle=False,
    num_workers=NUM_WORKERS_CORRUPT,
    pin_memory=PIN_MEMORY,
)
loader_val_bright = DataLoader(
    val_bright,
    batch_size=BATCH_SIZE,
    shuffle=False,
    num_workers=NUM_WORKERS_CORRUPT,
    pin_memory=PIN_MEMORY,
)
```

```
print("Loaders ready:", len(loader_val_clean), len(loader_val_blur), len(loader_val
print("workers clean/corrupt:", NUM_WORKERS_CLEAN, NUM_WORKERS_CORRUPT)
```

```
Loaders ready: 125 125 125 125
workers clean/corrupt: 4 0
```

Evaluate mIoU across conditions

In [7]:

```
import time
import torch

print("torch.cuda.is_available():", torch.cuda.is_available())
if torch.cuda.is_available():
    print("GPU:", torch.cuda.get_device_name(0))

print("device variable:", device)
print("model parameter device:", next(model.parameters()).device)

x, y, _ = next(iter(loader_val_clean))
print("batch (before to):", x.device, y.device, x.shape)

if torch.cuda.is_available():
    torch.cuda.empty_cache()
    torch.cuda.reset_peak_memory_stats()

t0 = time.time()
x = x.to(device, non_blocking=True)
with torch.inference_mode():
    out = model(x)["out"]
if torch.cuda.is_available():
    torch.cuda.synchronize()
t1 = time.time()

print("batch (after to):", x.device)
print("output device:", out.device, "shape:", tuple(out.shape))
print(f"one forward pass time: {t1 - t0:.3f}s")
if torch.cuda.is_available():
    print(f"cuda mem allocated: {torch.cuda.memory_allocated() / 1e9:.2f} GB")
    print(f"cuda peak allocated: {torch.cuda.max_memory_allocated() / 1e9:.2f} GB")
```

```
torch.cuda.is_available(): True
GPU: NVIDIA GeForce RTX 5070
device variable: cuda
model parameter device: cuda:0
batch (before to): cpu cpu torch.Size([4, 3, 512, 1024])
batch (after to): cuda:0
output device: cuda:0 shape: (4, 19, 512, 1024)
one forward pass time: 1.108s
cuda mem allocated: 0.53 GB
cuda peak allocated: 1.31 GB
```

In [8]:

```
import time
import importlib
import src.train_utils as train_utils
```

```python
# Pull latest evaluate() implementation from src while iterating in notebooks.
importlib.reload(train_utils)
evaluate = train_utils.evaluate

USE_AMP_EVAL = True
PROGRESS_EVERY = 1

print("Evaluation mode: full validation set")

results = {}
t0 = time.time()

for name, loader in [
    ("clean", loader_val_clean),
    ("blur", loader_val_blur),
    ("noise", loader_val_noise),
    ("bright", loader_val_bright),
]:
    t_cond = time.time()
    print(f"\nStarting {name}...")
    miou, _ = evaluate(
        model, loader, num_classes=19, device=device,
        use_amp=USE_AMP_EVAL, max_batches=None,
        progress=True, progress_every=PROGRESS_EVERY, progress_prefix=name,
    )
    results[name] = miou
    print(f"{name:>7s}: {miou:.4f}  ({time.time() - t_cond:.1f}s)")

elapsed = time.time() - t0
print(f"Total evaluation time: {elapsed/60:.2f} min")
results
```

```
Evaluation mode: full validation set

Starting clean...
[clean] 1/125 (  0.8%) | elapsed 0.8s | eta 102.9s
[clean] 2/125 (  1.6%) | elapsed 0.9s | eta 57.3s
[clean] 3/125 (  2.4%) | elapsed 1.0s | eta 42.5s
[clean] 4/125 (  3.2%) | elapsed 1.2s | eta 35.0s
[clean] 5/125 (  4.0%) | elapsed 1.3s | eta 30.2s
[clean] 6/125 (  4.8%) | elapsed 1.4s | eta 26.9s
[clean] 7/125 (  5.6%) | elapsed 1.5s | eta 24.6s
[clean] 8/125 (  6.4%) | elapsed 1.6s | eta 23.0s
[clean] 9/125 (  7.2%) | elapsed 1.7s | eta 21.6s
[clean] 10/125 (  8.0%) | elapsed 1.8s | eta 20.3s
[clean] 11/125 (  8.8%) | elapsed 1.9s | eta 19.3s
[clean] 12/125 (  9.6%) | elapsed 2.0s | eta 18.4s
[clean] 13/125 ( 10.4%) | elapsed 2.1s | eta 18.2s
[clean] 14/125 ( 11.2%) | elapsed 2.2s | eta 17.5s
[clean] 15/125 ( 12.0%) | elapsed 2.3s | eta 17.0s
[clean] 16/125 ( 12.8%) | elapsed 2.4s | eta 16.5s
[clean] 17/125 ( 13.6%) | elapsed 2.6s | eta 16.2s
[clean] 18/125 ( 14.4%) | elapsed 2.7s | eta 15.9s
[clean] 19/125 ( 15.2%) | elapsed 2.8s | eta 15.5s
[clean] 20/125 ( 16.0%) | elapsed 2.9s | eta 15.2s
[clean] 21/125 ( 16.8%) | elapsed 3.0s | eta 14.9s
[clean] 22/125 ( 17.6%) | elapsed 3.1s | eta 14.7s
[clean] 23/125 ( 18.4%) | elapsed 3.2s | eta 14.4s
[clean] 24/125 ( 19.2%) | elapsed 3.4s | eta 14.2s
[clean] 25/125 ( 20.0%) | elapsed 3.5s | eta 14.0s
[clean] 26/125 ( 20.8%) | elapsed 3.6s | eta 13.7s
[clean] 27/125 ( 21.6%) | elapsed 3.7s | eta 13.5s
[clean] 28/125 ( 22.4%) | elapsed 3.8s | eta 13.2s
[clean] 29/125 ( 23.2%) | elapsed 4.0s | eta 13.1s
[clean] 30/125 ( 24.0%) | elapsed 4.1s | eta 12.8s
[clean] 31/125 ( 24.8%) | elapsed 4.2s | eta 12.6s
[clean] 32/125 ( 25.6%) | elapsed 4.3s | eta 12.4s
[clean] 33/125 ( 26.4%) | elapsed 4.4s | eta 12.4s
[clean] 34/125 ( 27.2%) | elapsed 4.5s | eta 12.1s
[clean] 35/125 ( 28.0%) | elapsed 4.6s | eta 11.9s
[clean] 36/125 ( 28.8%) | elapsed 4.7s | eta 11.7s
[clean] 37/125 ( 29.6%) | elapsed 4.9s | eta 11.6s
[clean] 38/125 ( 30.4%) | elapsed 5.0s | eta 11.4s
[clean] 39/125 ( 31.2%) | elapsed 5.1s | eta 11.2s
[clean] 40/125 ( 32.0%) | elapsed 5.2s | eta 11.0s
[clean] 41/125 ( 32.8%) | elapsed 5.5s | eta 11.2s
[clean] 42/125 ( 33.6%) | elapsed 5.6s | eta 11.0s
[clean] 43/125 ( 34.4%) | elapsed 5.7s | eta 10.9s
[clean] 44/125 ( 35.2%) | elapsed 5.8s | eta 10.7s
[clean] 45/125 ( 36.0%) | elapsed 5.9s | eta 10.6s
[clean] 46/125 ( 36.8%) | elapsed 6.0s | eta 10.4s
[clean] 47/125 ( 37.6%) | elapsed 6.1s | eta 10.2s
[clean] 48/125 ( 38.4%) | elapsed 6.2s | eta 10.0s
[clean] 49/125 ( 39.2%) | elapsed 6.4s | eta 9.9s
[clean] 50/125 ( 40.0%) | elapsed 6.5s | eta 9.8s
[clean] 51/125 ( 40.8%) | elapsed 6.6s | eta 9.6s
[clean] 52/125 ( 41.6%) | elapsed 6.7s | eta 9.4s
[clean] 53/125 ( 42.4%) | elapsed 6.8s | eta 9.3s
```

```
[clean] 54/125 ( 43.2%) | elapsed 6.9s | eta 9.1s
[clean] 55/125 ( 44.0%) | elapsed 7.1s | eta 9.0s
[clean] 56/125 ( 44.8%) | elapsed 7.2s | eta 8.8s
[clean] 57/125 ( 45.6%) | elapsed 7.3s | eta 8.7s
[clean] 58/125 ( 46.4%) | elapsed 7.4s | eta 8.5s
[clean] 59/125 ( 47.2%) | elapsed 7.5s | eta 8.4s
[clean] 60/125 ( 48.0%) | elapsed 7.6s | eta 8.2s
[clean] 61/125 ( 48.8%) | elapsed 7.7s | eta 8.1s
[clean] 62/125 ( 49.6%) | elapsed 7.8s | eta 8.0s
[clean] 63/125 ( 50.4%) | elapsed 7.9s | eta 7.8s
[clean] 64/125 ( 51.2%) | elapsed 8.0s | eta 7.7s
[clean] 65/125 ( 52.0%) | elapsed 8.2s | eta 7.5s
[clean] 66/125 ( 52.8%) | elapsed 8.3s | eta 7.4s
[clean] 67/125 ( 53.6%) | elapsed 8.4s | eta 7.3s
[clean] 68/125 ( 54.4%) | elapsed 8.5s | eta 7.1s
[clean] 69/125 ( 55.2%) | elapsed 8.6s | eta 7.0s
[clean] 70/125 ( 56.0%) | elapsed 8.8s | eta 6.9s
[clean] 71/125 ( 56.8%) | elapsed 8.9s | eta 6.7s
[clean] 72/125 ( 57.6%) | elapsed 9.0s | eta 6.6s
[clean] 73/125 ( 58.4%) | elapsed 9.1s | eta 6.5s
[clean] 74/125 ( 59.2%) | elapsed 9.2s | eta 6.3s
[clean] 75/125 ( 60.0%) | elapsed 9.3s | eta 6.2s
[clean] 76/125 ( 60.8%) | elapsed 9.4s | eta 6.1s
[clean] 77/125 ( 61.6%) | elapsed 9.5s | eta 5.9s
[clean] 78/125 ( 62.4%) | elapsed 9.6s | eta 5.8s
[clean] 79/125 ( 63.2%) | elapsed 9.7s | eta 5.7s
[clean] 80/125 ( 64.0%) | elapsed 9.8s | eta 5.5s
[clean] 81/125 ( 64.8%) | elapsed 10.0s | eta 5.4s
[clean] 82/125 ( 65.6%) | elapsed 10.0s | eta 5.3s
[clean] 83/125 ( 66.4%) | elapsed 10.1s | eta 5.1s
[clean] 84/125 ( 67.2%) | elapsed 10.2s | eta 5.0s
[clean] 85/125 ( 68.0%) | elapsed 10.4s | eta 4.9s
[clean] 86/125 ( 68.8%) | elapsed 10.5s | eta 4.8s
[clean] 87/125 ( 69.6%) | elapsed 10.6s | eta 4.6s
[clean] 88/125 ( 70.4%) | elapsed 10.7s | eta 4.5s
[clean] 89/125 ( 71.2%) | elapsed 10.9s | eta 4.4s
[clean] 90/125 ( 72.0%) | elapsed 11.0s | eta 4.3s
[clean] 91/125 ( 72.8%) | elapsed 11.0s | eta 4.1s
[clean] 92/125 ( 73.6%) | elapsed 11.1s | eta 4.0s
[clean] 93/125 ( 74.4%) | elapsed 11.3s | eta 3.9s
[clean] 94/125 ( 75.2%) | elapsed 11.4s | eta 3.8s
[clean] 95/125 ( 76.0%) | elapsed 11.5s | eta 3.6s
[clean] 96/125 ( 76.8%) | elapsed 11.6s | eta 3.5s
[clean] 97/125 ( 77.6%) | elapsed 11.9s | eta 3.4s
[clean] 98/125 ( 78.4%) | elapsed 11.9s | eta 3.3s
[clean] 99/125 ( 79.2%) | elapsed 12.0s | eta 3.2s
[clean] 100/125 ( 80.0%) | elapsed 12.1s | eta 3.0s
[clean] 101/125 ( 80.8%) | elapsed 12.3s | eta 2.9s
[clean] 102/125 ( 81.6%) | elapsed 12.4s | eta 2.8s
[clean] 103/125 ( 82.4%) | elapsed 12.5s | eta 2.7s
[clean] 104/125 ( 83.2%) | elapsed 12.5s | eta 2.5s
[clean] 105/125 ( 84.0%) | elapsed 12.7s | eta 2.4s
[clean] 106/125 ( 84.8%) | elapsed 12.8s | eta 2.3s
[clean] 107/125 ( 85.6%) | elapsed 13.0s | eta 2.2s
[clean] 108/125 ( 86.4%) | elapsed 13.0s | eta 2.1s
[clean] 109/125 ( 87.2%) | elapsed 13.2s | eta 1.9s
```

```
[clean] 110/125 ( 88.0%) | elapsed 13.3s | eta 1.8s
[clean] 111/125 ( 88.8%) | elapsed 13.4s | eta 1.7s
[clean] 112/125 ( 89.6%) | elapsed 13.4s | eta 1.6s
[clean] 113/125 ( 90.4%) | elapsed 13.6s | eta 1.4s
[clean] 114/125 ( 91.2%) | elapsed 13.7s | eta 1.3s
[clean] 115/125 ( 92.0%) | elapsed 13.8s | eta 1.2s
[clean] 116/125 ( 92.8%) | elapsed 13.9s | eta 1.1s
[clean] 117/125 ( 93.6%) | elapsed 14.1s | eta 1.0s
[clean] 118/125 ( 94.4%) | elapsed 14.2s | eta 0.8s
[clean] 119/125 ( 95.2%) | elapsed 14.3s | eta 0.7s
[clean] 120/125 ( 96.0%) | elapsed 14.4s | eta 0.6s
[clean] 121/125 ( 96.8%) | elapsed 14.5s | eta 0.5s
[clean] 122/125 ( 97.6%) | elapsed 14.6s | eta 0.4s
[clean] 123/125 ( 98.4%) | elapsed 14.7s | eta 0.2s
[clean] 124/125 ( 99.2%) | elapsed 14.8s | eta 0.1s
[clean] 125/125 (100.0%) | elapsed 14.9s | eta 0.0s
  clean: 0.6483  (14.9s)

Starting blur...
[blur] 1/125 (  0.8%) | elapsed 0.4s | eta 49.5s
[blur] 2/125 (  1.6%) | elapsed 0.8s | eta 46.9s
[blur] 3/125 (  2.4%) | elapsed 1.1s | eta 45.7s
[blur] 4/125 (  3.2%) | elapsed 1.5s | eta 44.6s
[blur] 5/125 (  4.0%) | elapsed 1.8s | eta 43.5s
[blur] 6/125 (  4.8%) | elapsed 2.1s | eta 42.5s
[blur] 7/125 (  5.6%) | elapsed 2.5s | eta 41.8s
[blur] 8/125 (  6.4%) | elapsed 2.8s | eta 41.4s
[blur] 9/125 (  7.2%) | elapsed 3.2s | eta 41.1s
[blur] 10/125 (  8.0%) | elapsed 3.5s | eta 40.6s
[blur] 11/125 (  8.8%) | elapsed 3.9s | eta 40.1s
[blur] 12/125 (  9.6%) | elapsed 4.2s | eta 39.8s
[blur] 13/125 ( 10.4%) | elapsed 4.6s | eta 39.4s
[blur] 14/125 ( 11.2%) | elapsed 4.9s | eta 39.0s
[blur] 15/125 ( 12.0%) | elapsed 5.3s | eta 38.6s
[blur] 16/125 ( 12.8%) | elapsed 5.6s | eta 38.2s
[blur] 17/125 ( 13.6%) | elapsed 6.0s | eta 37.8s
[blur] 18/125 ( 14.4%) | elapsed 6.3s | eta 37.4s
[blur] 19/125 ( 15.2%) | elapsed 6.6s | eta 37.1s
[blur] 20/125 ( 16.0%) | elapsed 7.0s | eta 36.8s
[blur] 21/125 ( 16.8%) | elapsed 7.4s | eta 36.4s
[blur] 22/125 ( 17.6%) | elapsed 7.7s | eta 36.1s
[blur] 23/125 ( 18.4%) | elapsed 8.1s | eta 35.8s
[blur] 24/125 ( 19.2%) | elapsed 8.4s | eta 35.5s
[blur] 25/125 ( 20.0%) | elapsed 8.8s | eta 35.2s
[blur] 26/125 ( 20.8%) | elapsed 9.2s | eta 34.9s
[blur] 27/125 ( 21.6%) | elapsed 9.5s | eta 34.5s
[blur] 28/125 ( 22.4%) | elapsed 9.9s | eta 34.1s
[blur] 29/125 ( 23.2%) | elapsed 10.2s | eta 33.8s
[blur] 30/125 ( 24.0%) | elapsed 10.5s | eta 33.4s
[blur] 31/125 ( 24.8%) | elapsed 10.9s | eta 33.0s
[blur] 32/125 ( 25.6%) | elapsed 11.2s | eta 32.6s
[blur] 33/125 ( 26.4%) | elapsed 11.6s | eta 32.2s
[blur] 34/125 ( 27.2%) | elapsed 11.9s | eta 31.9s
[blur] 35/125 ( 28.0%) | elapsed 12.3s | eta 31.5s
[blur] 36/125 ( 28.8%) | elapsed 12.6s | eta 31.2s
[blur] 37/125 ( 29.6%) | elapsed 12.9s | eta 30.8s
```

```
[blur] 38/125 ( 30.4%) | elapsed 13.3s | eta 30.4s
[blur] 39/125 ( 31.2%) | elapsed 13.6s | eta 30.1s
[blur] 40/125 ( 32.0%) | elapsed 14.0s | eta 29.7s
[blur] 41/125 ( 32.8%) | elapsed 14.3s | eta 29.4s
[blur] 42/125 ( 33.6%) | elapsed 14.7s | eta 29.0s
[blur] 43/125 ( 34.4%) | elapsed 15.0s | eta 28.6s
[blur] 44/125 ( 35.2%) | elapsed 15.4s | eta 28.3s
[blur] 45/125 ( 36.0%) | elapsed 15.7s | eta 27.9s
[blur] 46/125 ( 36.8%) | elapsed 16.0s | eta 27.5s
[blur] 47/125 ( 37.6%) | elapsed 16.4s | eta 27.2s
[blur] 48/125 ( 38.4%) | elapsed 16.7s | eta 26.8s
[blur] 49/125 ( 39.2%) | elapsed 17.1s | eta 26.5s
[blur] 50/125 ( 40.0%) | elapsed 17.4s | eta 26.1s
[blur] 51/125 ( 40.8%) | elapsed 17.7s | eta 25.7s
[blur] 52/125 ( 41.6%) | elapsed 18.1s | eta 25.4s
[blur] 53/125 ( 42.4%) | elapsed 18.4s | eta 25.0s
[blur] 54/125 ( 43.2%) | elapsed 18.8s | eta 24.7s
[blur] 55/125 ( 44.0%) | elapsed 19.1s | eta 24.3s
[blur] 56/125 ( 44.8%) | elapsed 19.4s | eta 23.9s
[blur] 57/125 ( 45.6%) | elapsed 19.8s | eta 23.6s
[blur] 58/125 ( 46.4%) | elapsed 20.1s | eta 23.2s
[blur] 59/125 ( 47.2%) | elapsed 20.5s | eta 22.9s
[blur] 60/125 ( 48.0%) | elapsed 20.8s | eta 22.5s
[blur] 61/125 ( 48.8%) | elapsed 21.2s | eta 22.2s
[blur] 62/125 ( 49.6%) | elapsed 21.5s | eta 21.9s
[blur] 63/125 ( 50.4%) | elapsed 21.9s | eta 21.5s
[blur] 64/125 ( 51.2%) | elapsed 22.2s | eta 21.2s
[blur] 65/125 ( 52.0%) | elapsed 22.6s | eta 20.8s
[blur] 66/125 ( 52.8%) | elapsed 22.9s | eta 20.5s
[blur] 67/125 ( 53.6%) | elapsed 23.3s | eta 20.2s
[blur] 68/125 ( 54.4%) | elapsed 23.7s | eta 19.9s
[blur] 69/125 ( 55.2%) | elapsed 24.0s | eta 19.5s
[blur] 70/125 ( 56.0%) | elapsed 24.4s | eta 19.2s
[blur] 71/125 ( 56.8%) | elapsed 24.8s | eta 18.8s
[blur] 72/125 ( 57.6%) | elapsed 25.1s | eta 18.5s
[blur] 73/125 ( 58.4%) | elapsed 25.4s | eta 18.1s
[blur] 74/125 ( 59.2%) | elapsed 25.8s | eta 17.8s
[blur] 75/125 ( 60.0%) | elapsed 26.1s | eta 17.4s
[blur] 76/125 ( 60.8%) | elapsed 26.4s | eta 17.0s
[blur] 77/125 ( 61.6%) | elapsed 26.8s | eta 16.7s
[blur] 78/125 ( 62.4%) | elapsed 27.1s | eta 16.3s
[blur] 79/125 ( 63.2%) | elapsed 27.5s | eta 16.0s
[blur] 80/125 ( 64.0%) | elapsed 27.8s | eta 15.6s
[blur] 81/125 ( 64.8%) | elapsed 28.1s | eta 15.3s
[blur] 82/125 ( 65.6%) | elapsed 28.5s | eta 14.9s
[blur] 83/125 ( 66.4%) | elapsed 28.8s | eta 14.6s
[blur] 84/125 ( 67.2%) | elapsed 29.1s | eta 14.2s
[blur] 85/125 ( 68.0%) | elapsed 29.5s | eta 13.9s
[blur] 86/125 ( 68.8%) | elapsed 29.8s | eta 13.5s
[blur] 87/125 ( 69.6%) | elapsed 30.1s | eta 13.2s
[blur] 88/125 ( 70.4%) | elapsed 30.5s | eta 12.8s
[blur] 89/125 ( 71.2%) | elapsed 30.8s | eta 12.5s
[blur] 90/125 ( 72.0%) | elapsed 31.1s | eta 12.1s
[blur] 91/125 ( 72.8%) | elapsed 31.5s | eta 11.8s
[blur] 92/125 ( 73.6%) | elapsed 31.8s | eta 11.4s
[blur] 93/125 ( 74.4%) | elapsed 32.1s | eta 11.1s
```

```
[blur] 94/125 ( 75.2%) | elapsed 32.5s | eta 10.7s
[blur] 95/125 ( 76.0%) | elapsed 32.8s | eta 10.4s
[blur] 96/125 ( 76.8%) | elapsed 33.2s | eta 10.0s
[blur] 97/125 ( 77.6%) | elapsed 33.5s | eta 9.7s
[blur] 98/125 ( 78.4%) | elapsed 33.8s | eta 9.3s
[blur] 99/125 ( 79.2%) | elapsed 34.1s | eta 9.0s
[blur] 100/125 ( 80.0%) | elapsed 34.5s | eta 8.6s
[blur] 101/125 ( 80.8%) | elapsed 34.8s | eta 8.3s
[blur] 102/125 ( 81.6%) | elapsed 35.1s | eta 7.9s
[blur] 103/125 ( 82.4%) | elapsed 35.5s | eta 7.6s
[blur] 104/125 ( 83.2%) | elapsed 35.8s | eta 7.2s
[blur] 105/125 ( 84.0%) | elapsed 36.1s | eta 6.9s
[blur] 106/125 ( 84.8%) | elapsed 36.5s | eta 6.5s
[blur] 107/125 ( 85.6%) | elapsed 36.8s | eta 6.2s
[blur] 108/125 ( 86.4%) | elapsed 37.1s | eta 5.8s
[blur] 109/125 ( 87.2%) | elapsed 37.5s | eta 5.5s
[blur] 110/125 ( 88.0%) | elapsed 37.8s | eta 5.2s
[blur] 111/125 ( 88.8%) | elapsed 38.1s | eta 4.8s
[blur] 112/125 ( 89.6%) | elapsed 38.5s | eta 4.5s
[blur] 113/125 ( 90.4%) | elapsed 38.8s | eta 4.1s
[blur] 114/125 ( 91.2%) | elapsed 39.1s | eta 3.8s
[blur] 115/125 ( 92.0%) | elapsed 39.5s | eta 3.4s
[blur] 116/125 ( 92.8%) | elapsed 39.8s | eta 3.1s
[blur] 117/125 ( 93.6%) | elapsed 40.1s | eta 2.7s
[blur] 118/125 ( 94.4%) | elapsed 40.5s | eta 2.4s
[blur] 119/125 ( 95.2%) | elapsed 40.8s | eta 2.1s
[blur] 120/125 ( 96.0%) | elapsed 41.1s | eta 1.7s
[blur] 121/125 ( 96.8%) | elapsed 41.4s | eta 1.4s
[blur] 122/125 ( 97.6%) | elapsed 41.8s | eta 1.0s
[blur] 123/125 ( 98.4%) | elapsed 42.1s | eta 0.7s
[blur] 124/125 ( 99.2%) | elapsed 42.5s | eta 0.3s
[blur] 125/125 (100.0%) | elapsed 42.9s | eta 0.0s
   blur: 0.5285  (42.9s)

Starting noise...
[noise] 1/125 (  0.8%) | elapsed 0.4s | eta 43.7s
[noise] 2/125 (  1.6%) | elapsed 0.7s | eta 43.2s
[noise] 3/125 (  2.4%) | elapsed 1.1s | eta 42.9s
[noise] 4/125 (  3.2%) | elapsed 1.4s | eta 42.4s
[noise] 5/125 (  4.0%) | elapsed 1.8s | eta 42.2s
[noise] 6/125 (  4.8%) | elapsed 2.1s | eta 42.3s
[noise] 7/125 (  5.6%) | elapsed 2.5s | eta 42.0s
[noise] 8/125 (  6.4%) | elapsed 2.9s | eta 41.8s
[noise] 9/125 (  7.2%) | elapsed 3.2s | eta 41.4s
[noise] 10/125 (  8.0%) | elapsed 3.6s | eta 41.1s
[noise] 11/125 (  8.8%) | elapsed 3.9s | eta 40.8s
[noise] 12/125 (  9.6%) | elapsed 4.3s | eta 40.5s
[noise] 13/125 ( 10.4%) | elapsed 4.7s | eta 40.3s
[noise] 14/125 ( 11.2%) | elapsed 5.0s | eta 40.0s
[noise] 15/125 ( 12.0%) | elapsed 5.4s | eta 39.5s
[noise] 16/125 ( 12.8%) | elapsed 5.7s | eta 39.2s
[noise] 17/125 ( 13.6%) | elapsed 6.1s | eta 38.8s
[noise] 18/125 ( 14.4%) | elapsed 6.5s | eta 38.5s
[noise] 19/125 ( 15.2%) | elapsed 6.8s | eta 38.0s
[noise] 20/125 ( 16.0%) | elapsed 7.2s | eta 37.7s
[noise] 21/125 ( 16.8%) | elapsed 7.5s | eta 37.2s
```

```
[noise] 22/125 ( 17.6%) | elapsed 7.9s | eta 36.8s
[noise] 23/125 ( 18.4%) | elapsed 8.2s | eta 36.4s
[noise] 24/125 ( 19.2%) | elapsed 8.6s | eta 36.0s
[noise] 25/125 ( 20.0%) | elapsed 8.9s | eta 35.6s
[noise] 26/125 ( 20.8%) | elapsed 9.3s | eta 35.2s
[noise] 27/125 ( 21.6%) | elapsed 9.6s | eta 34.9s
[noise] 28/125 ( 22.4%) | elapsed 10.0s | eta 34.5s
[noise] 29/125 ( 23.2%) | elapsed 10.3s | eta 34.1s
[noise] 30/125 ( 24.0%) | elapsed 10.7s | eta 33.7s
[noise] 31/125 ( 24.8%) | elapsed 11.0s | eta 33.4s
[noise] 32/125 ( 25.6%) | elapsed 11.3s | eta 33.0s
[noise] 33/125 ( 26.4%) | elapsed 11.7s | eta 32.6s
[noise] 34/125 ( 27.2%) | elapsed 12.0s | eta 32.2s
[noise] 35/125 ( 28.0%) | elapsed 12.4s | eta 31.9s
[noise] 36/125 ( 28.8%) | elapsed 12.7s | eta 31.5s
[noise] 37/125 ( 29.6%) | elapsed 13.1s | eta 31.1s
[noise] 38/125 ( 30.4%) | elapsed 13.4s | eta 30.8s
[noise] 39/125 ( 31.2%) | elapsed 13.8s | eta 30.4s
[noise] 40/125 ( 32.0%) | elapsed 14.1s | eta 30.0s
[noise] 41/125 ( 32.8%) | elapsed 14.5s | eta 29.7s
[noise] 42/125 ( 33.6%) | elapsed 14.8s | eta 29.3s
[noise] 43/125 ( 34.4%) | elapsed 15.2s | eta 29.0s
[noise] 44/125 ( 35.2%) | elapsed 15.5s | eta 28.6s
[noise] 45/125 ( 36.0%) | elapsed 15.9s | eta 28.2s
[noise] 46/125 ( 36.8%) | elapsed 16.2s | eta 27.9s
[noise] 47/125 ( 37.6%) | elapsed 16.6s | eta 27.5s
[noise] 48/125 ( 38.4%) | elapsed 16.9s | eta 27.1s
[noise] 49/125 ( 39.2%) | elapsed 17.3s | eta 26.8s
[noise] 50/125 ( 40.0%) | elapsed 17.6s | eta 26.4s
[noise] 51/125 ( 40.8%) | elapsed 17.9s | eta 26.0s
[noise] 52/125 ( 41.6%) | elapsed 18.3s | eta 25.7s
[noise] 53/125 ( 42.4%) | elapsed 18.6s | eta 25.3s
[noise] 54/125 ( 43.2%) | elapsed 19.0s | eta 25.0s
[noise] 55/125 ( 44.0%) | elapsed 19.3s | eta 24.6s
[noise] 56/125 ( 44.8%) | elapsed 19.7s | eta 24.2s
[noise] 57/125 ( 45.6%) | elapsed 20.0s | eta 23.9s
[noise] 58/125 ( 46.4%) | elapsed 20.4s | eta 23.5s
[noise] 59/125 ( 47.2%) | elapsed 20.7s | eta 23.2s
[noise] 60/125 ( 48.0%) | elapsed 21.1s | eta 22.8s
[noise] 61/125 ( 48.8%) | elapsed 21.4s | eta 22.5s
[noise] 62/125 ( 49.6%) | elapsed 21.8s | eta 22.1s
[noise] 63/125 ( 50.4%) | elapsed 22.1s | eta 21.8s
[noise] 64/125 ( 51.2%) | elapsed 22.5s | eta 21.4s
[noise] 65/125 ( 52.0%) | elapsed 22.8s | eta 21.1s
[noise] 66/125 ( 52.8%) | elapsed 23.2s | eta 20.7s
[noise] 67/125 ( 53.6%) | elapsed 23.5s | eta 20.4s
[noise] 68/125 ( 54.4%) | elapsed 23.8s | eta 20.0s
[noise] 69/125 ( 55.2%) | elapsed 24.2s | eta 19.6s
[noise] 70/125 ( 56.0%) | elapsed 24.5s | eta 19.3s
[noise] 71/125 ( 56.8%) | elapsed 24.9s | eta 18.9s
[noise] 72/125 ( 57.6%) | elapsed 25.2s | eta 18.6s
[noise] 73/125 ( 58.4%) | elapsed 25.6s | eta 18.2s
[noise] 74/125 ( 59.2%) | elapsed 25.9s | eta 17.9s
[noise] 75/125 ( 60.0%) | elapsed 26.3s | eta 17.5s
[noise] 76/125 ( 60.8%) | elapsed 26.6s | eta 17.2s
[noise] 77/125 ( 61.6%) | elapsed 26.9s | eta 16.8s
```

```
[noise] 78/125 ( 62.4%) | elapsed 27.3s | eta 16.4s
[noise] 79/125 ( 63.2%) | elapsed 27.6s | eta 16.1s
[noise] 80/125 ( 64.0%) | elapsed 28.0s | eta 15.7s
[noise] 81/125 ( 64.8%) | elapsed 28.3s | eta 15.4s
[noise] 82/125 ( 65.6%) | elapsed 28.7s | eta 15.0s
[noise] 83/125 ( 66.4%) | elapsed 29.0s | eta 14.7s
[noise] 84/125 ( 67.2%) | elapsed 29.4s | eta 14.3s
[noise] 85/125 ( 68.0%) | elapsed 29.7s | eta 14.0s
[noise] 86/125 ( 68.8%) | elapsed 30.1s | eta 13.6s
[noise] 87/125 ( 69.6%) | elapsed 30.4s | eta 13.3s
[noise] 88/125 ( 70.4%) | elapsed 30.7s | eta 12.9s
[noise] 89/125 ( 71.2%) | elapsed 31.1s | eta 12.6s
[noise] 90/125 ( 72.0%) | elapsed 31.4s | eta 12.2s
[noise] 91/125 ( 72.8%) | elapsed 31.8s | eta 11.9s
[noise] 92/125 ( 73.6%) | elapsed 32.1s | eta 11.5s
[noise] 93/125 ( 74.4%) | elapsed 32.5s | eta 11.2s
[noise] 94/125 ( 75.2%) | elapsed 32.8s | eta 10.8s
[noise] 95/125 ( 76.0%) | elapsed 33.2s | eta 10.5s
[noise] 96/125 ( 76.8%) | elapsed 33.5s | eta 10.1s
[noise] 97/125 ( 77.6%) | elapsed 33.9s | eta 9.8s
[noise] 98/125 ( 78.4%) | elapsed 34.2s | eta 9.4s
[noise] 99/125 ( 79.2%) | elapsed 34.5s | eta 9.1s
[noise] 100/125 ( 80.0%) | elapsed 34.9s | eta 8.7s
[noise] 101/125 ( 80.8%) | elapsed 35.2s | eta 8.4s
[noise] 102/125 ( 81.6%) | elapsed 35.6s | eta 8.0s
[noise] 103/125 ( 82.4%) | elapsed 35.9s | eta 7.7s
[noise] 104/125 ( 83.2%) | elapsed 36.3s | eta 7.3s
[noise] 105/125 ( 84.0%) | elapsed 36.7s | eta 7.0s
[noise] 106/125 ( 84.8%) | elapsed 37.0s | eta 6.6s
[noise] 107/125 ( 85.6%) | elapsed 37.4s | eta 6.3s
[noise] 108/125 ( 86.4%) | elapsed 37.8s | eta 5.9s
[noise] 109/125 ( 87.2%) | elapsed 38.2s | eta 5.6s
[noise] 110/125 ( 88.0%) | elapsed 38.5s | eta 5.3s
[noise] 111/125 ( 88.8%) | elapsed 38.9s | eta 4.9s
[noise] 112/125 ( 89.6%) | elapsed 39.2s | eta 4.5s
[noise] 113/125 ( 90.4%) | elapsed 39.6s | eta 4.2s
[noise] 114/125 ( 91.2%) | elapsed 39.9s | eta 3.9s
[noise] 115/125 ( 92.0%) | elapsed 40.3s | eta 3.5s
[noise] 116/125 ( 92.8%) | elapsed 40.6s | eta 3.2s
[noise] 117/125 ( 93.6%) | elapsed 41.0s | eta 2.8s
[noise] 118/125 ( 94.4%) | elapsed 41.3s | eta 2.5s
[noise] 119/125 ( 95.2%) | elapsed 41.7s | eta 2.1s
[noise] 120/125 ( 96.0%) | elapsed 42.0s | eta 1.8s
[noise] 121/125 ( 96.8%) | elapsed 42.4s | eta 1.4s
[noise] 122/125 ( 97.6%) | elapsed 42.7s | eta 1.1s
[noise] 123/125 ( 98.4%) | elapsed 43.1s | eta 0.7s
[noise] 124/125 ( 99.2%) | elapsed 43.4s | eta 0.4s
[noise] 125/125 (100.0%) | elapsed 43.7s | eta 0.0s
  noise: 0.3197  (43.7s)

Starting bright...
[bright] 1/125 (  0.8%) | elapsed 0.3s | eta 39.4s
[bright] 2/125 (  1.6%) | elapsed 0.6s | eta 39.1s
[bright] 3/125 (  2.4%) | elapsed 0.9s | eta 38.6s
[bright] 4/125 (  3.2%) | elapsed 1.3s | eta 38.2s
[bright] 5/125 (  4.0%) | elapsed 1.6s | eta 37.8s
```

```
[bright] 6/125 (  4.8%) | elapsed 1.9s | eta 37.4s
[bright] 7/125 (  5.6%) | elapsed 2.2s | eta 37.1s
[bright] 8/125 (  6.4%) | elapsed 2.5s | eta 36.8s
[bright] 9/125 (  7.2%) | elapsed 2.8s | eta 36.5s
[bright] 10/125 (  8.0%) | elapsed 3.2s | eta 36.3s
[bright] 11/125 (  8.8%) | elapsed 3.5s | eta 36.0s
[bright] 12/125 (  9.6%) | elapsed 3.8s | eta 35.7s
[bright] 13/125 ( 10.4%) | elapsed 4.1s | eta 35.4s
[bright] 14/125 ( 11.2%) | elapsed 4.4s | eta 35.1s
[bright] 15/125 ( 12.0%) | elapsed 4.7s | eta 34.8s
[bright] 16/125 ( 12.8%) | elapsed 5.1s | eta 34.5s
[bright] 17/125 ( 13.6%) | elapsed 5.4s | eta 34.2s
[bright] 18/125 ( 14.4%) | elapsed 5.7s | eta 33.9s
[bright] 19/125 ( 15.2%) | elapsed 6.0s | eta 33.5s
[bright] 20/125 ( 16.0%) | elapsed 6.3s | eta 33.2s
[bright] 21/125 ( 16.8%) | elapsed 6.6s | eta 32.8s
[bright] 22/125 ( 17.6%) | elapsed 7.0s | eta 32.6s
[bright] 23/125 ( 18.4%) | elapsed 7.3s | eta 32.3s
[bright] 24/125 ( 19.2%) | elapsed 7.6s | eta 32.0s
[bright] 25/125 ( 20.0%) | elapsed 7.9s | eta 31.7s
[bright] 26/125 ( 20.8%) | elapsed 8.2s | eta 31.3s
[bright] 27/125 ( 21.6%) | elapsed 8.5s | eta 31.0s
[bright] 28/125 ( 22.4%) | elapsed 8.9s | eta 30.7s
[bright] 29/125 ( 23.2%) | elapsed 9.2s | eta 30.4s
[bright] 30/125 ( 24.0%) | elapsed 9.5s | eta 30.1s
[bright] 31/125 ( 24.8%) | elapsed 9.8s | eta 29.7s
[bright] 32/125 ( 25.6%) | elapsed 10.1s | eta 29.4s
[bright] 33/125 ( 26.4%) | elapsed 10.4s | eta 29.1s
[bright] 34/125 ( 27.2%) | elapsed 10.7s | eta 28.7s
[bright] 35/125 ( 28.0%) | elapsed 11.0s | eta 28.4s
[bright] 36/125 ( 28.8%) | elapsed 11.4s | eta 28.1s
[bright] 37/125 ( 29.6%) | elapsed 11.7s | eta 27.8s
[bright] 38/125 ( 30.4%) | elapsed 12.0s | eta 27.5s
[bright] 39/125 ( 31.2%) | elapsed 12.3s | eta 27.2s
[bright] 40/125 ( 32.0%) | elapsed 12.6s | eta 26.9s
[bright] 41/125 ( 32.8%) | elapsed 13.0s | eta 26.5s
[bright] 42/125 ( 33.6%) | elapsed 13.3s | eta 26.2s
[bright] 43/125 ( 34.4%) | elapsed 13.6s | eta 25.9s
[bright] 44/125 ( 35.2%) | elapsed 13.9s | eta 25.6s
[bright] 45/125 ( 36.0%) | elapsed 14.2s | eta 25.3s
[bright] 46/125 ( 36.8%) | elapsed 14.5s | eta 25.0s
[bright] 47/125 ( 37.6%) | elapsed 14.9s | eta 24.7s
[bright] 48/125 ( 38.4%) | elapsed 15.2s | eta 24.3s
[bright] 49/125 ( 39.2%) | elapsed 15.5s | eta 24.0s
[bright] 50/125 ( 40.0%) | elapsed 15.8s | eta 23.7s
[bright] 51/125 ( 40.8%) | elapsed 16.1s | eta 23.4s
[bright] 52/125 ( 41.6%) | elapsed 16.4s | eta 23.0s
[bright] 53/125 ( 42.4%) | elapsed 16.7s | eta 22.7s
[bright] 54/125 ( 43.2%) | elapsed 17.1s | eta 22.4s
[bright] 55/125 ( 44.0%) | elapsed 17.4s | eta 22.1s
[bright] 56/125 ( 44.8%) | elapsed 17.7s | eta 21.8s
[bright] 57/125 ( 45.6%) | elapsed 18.0s | eta 21.5s
[bright] 58/125 ( 46.4%) | elapsed 18.3s | eta 21.2s
[bright] 59/125 ( 47.2%) | elapsed 18.6s | eta 20.8s
[bright] 60/125 ( 48.0%) | elapsed 19.0s | eta 20.5s
[bright] 61/125 ( 48.8%) | elapsed 19.3s | eta 20.2s
```

```
[bright] 62/125 ( 49.6%) | elapsed 19.6s | eta 19.9s
[bright] 63/125 ( 50.4%) | elapsed 19.9s | eta 19.6s
[bright] 64/125 ( 51.2%) | elapsed 20.2s | eta 19.3s
[bright] 65/125 ( 52.0%) | elapsed 20.5s | eta 19.0s
[bright] 66/125 ( 52.8%) | elapsed 20.9s | eta 18.6s
[bright] 67/125 ( 53.6%) | elapsed 21.2s | eta 18.3s
[bright] 68/125 ( 54.4%) | elapsed 21.5s | eta 18.0s
[bright] 69/125 ( 55.2%) | elapsed 21.8s | eta 17.7s
[bright] 70/125 ( 56.0%) | elapsed 22.1s | eta 17.4s
[bright] 71/125 ( 56.8%) | elapsed 22.4s | eta 17.0s
[bright] 72/125 ( 57.6%) | elapsed 22.7s | eta 16.7s
[bright] 73/125 ( 58.4%) | elapsed 23.0s | eta 16.4s
[bright] 74/125 ( 59.2%) | elapsed 23.4s | eta 16.1s
[bright] 75/125 ( 60.0%) | elapsed 23.7s | eta 15.8s
[bright] 76/125 ( 60.8%) | elapsed 24.0s | eta 15.5s
[bright] 77/125 ( 61.6%) | elapsed 24.3s | eta 15.1s
[bright] 78/125 ( 62.4%) | elapsed 24.6s | eta 14.8s
[bright] 79/125 ( 63.2%) | elapsed 24.9s | eta 14.5s
[bright] 80/125 ( 64.0%) | elapsed 25.2s | eta 14.2s
[bright] 81/125 ( 64.8%) | elapsed 25.6s | eta 13.9s
[bright] 82/125 ( 65.6%) | elapsed 25.9s | eta 13.6s
[bright] 83/125 ( 66.4%) | elapsed 26.2s | eta 13.2s
[bright] 84/125 ( 67.2%) | elapsed 26.5s | eta 12.9s
[bright] 85/125 ( 68.0%) | elapsed 26.8s | eta 12.6s
[bright] 86/125 ( 68.8%) | elapsed 27.1s | eta 12.3s
[bright] 87/125 ( 69.6%) | elapsed 27.4s | eta 12.0s
[bright] 88/125 ( 70.4%) | elapsed 27.7s | eta 11.7s
[bright] 89/125 ( 71.2%) | elapsed 28.1s | eta 11.4s
[bright] 90/125 ( 72.0%) | elapsed 28.4s | eta 11.0s
[bright] 91/125 ( 72.8%) | elapsed 28.7s | eta 10.7s
[bright] 92/125 ( 73.6%) | elapsed 29.0s | eta 10.4s
[bright] 93/125 ( 74.4%) | elapsed 29.3s | eta 10.1s
[bright] 94/125 ( 75.2%) | elapsed 29.6s | eta 9.8s
[bright] 95/125 ( 76.0%) | elapsed 30.0s | eta 9.5s
[bright] 96/125 ( 76.8%) | elapsed 30.3s | eta 9.1s
[bright] 97/125 ( 77.6%) | elapsed 30.6s | eta 8.8s
[bright] 98/125 ( 78.4%) | elapsed 30.9s | eta 8.5s
[bright] 99/125 ( 79.2%) | elapsed 31.2s | eta 8.2s
[bright] 100/125 ( 80.0%) | elapsed 31.5s | eta 7.9s
[bright] 101/125 ( 80.8%) | elapsed 31.8s | eta 7.6s
[bright] 102/125 ( 81.6%) | elapsed 32.1s | eta 7.2s
[bright] 103/125 ( 82.4%) | elapsed 32.4s | eta 6.9s
[bright] 104/125 ( 83.2%) | elapsed 32.7s | eta 6.6s
[bright] 105/125 ( 84.0%) | elapsed 33.1s | eta 6.3s
[bright] 106/125 ( 84.8%) | elapsed 33.4s | eta 6.0s
[bright] 107/125 ( 85.6%) | elapsed 33.7s | eta 5.7s
[bright] 108/125 ( 86.4%) | elapsed 34.0s | eta 5.4s
[bright] 109/125 ( 87.2%) | elapsed 34.3s | eta 5.0s
[bright] 110/125 ( 88.0%) | elapsed 34.6s | eta 4.7s
[bright] 111/125 ( 88.8%) | elapsed 34.9s | eta 4.4s
[bright] 112/125 ( 89.6%) | elapsed 35.2s | eta 4.1s
[bright] 113/125 ( 90.4%) | elapsed 35.5s | eta 3.8s
[bright] 114/125 ( 91.2%) | elapsed 35.8s | eta 3.5s
[bright] 115/125 ( 92.0%) | elapsed 36.2s | eta 3.1s
[bright] 116/125 ( 92.8%) | elapsed 36.5s | eta 2.8s
[bright] 117/125 ( 93.6%) | elapsed 36.8s | eta 2.5s
```

```
[bright] 118/125 ( 94.4%) | elapsed 37.1s | eta 2.2s
[bright] 119/125 ( 95.2%) | elapsed 37.4s | eta 1.9s
[bright] 120/125 ( 96.0%) | elapsed 37.7s | eta 1.6s
[bright] 121/125 ( 96.8%) | elapsed 38.0s | eta 1.3s
[bright] 122/125 ( 97.6%) | elapsed 38.3s | eta 0.9s
[bright] 123/125 ( 98.4%) | elapsed 38.6s | eta 0.6s
[bright] 124/125 ( 99.2%) | elapsed 39.0s | eta 0.3s
[bright] 125/125 (100.0%) | elapsed 39.3s | eta 0.0s
 bright: 0.5831  (39.3s)
Total evaluation time: 2.35 min
```

Out[8]:

```
{'clean': 0.6482956409454346,
 'blur': 0.5285175442695618,
 'noise': 0.31972113251686096,
 'bright': 0.583081066608429}
```

Print a clean results table

In [9]:

```python
print("mIoU results (val):")
for k in ["clean", "blur", "noise", "bright"]:
    print(f"{k:>7s}: {results[k]:.4f}")

with open(OUT / "shift_eval_results.txt", "w") as f:
    for k in ["clean", "blur", "noise", "bright"]:
        f.write(f"{k}: {results[k]:.6f}\n")
print("Saved:", OUT / "shift_eval_results.txt")
```

```
mIoU results (val):
  clean: 0.6483
   blur: 0.5285
  noise: 0.3197
 bright: 0.5831
Saved: c:\Users\dhruv\temporal_cityscapes\outputs\shift_eval_results.txt
```

Qualitative visualization utilities

In [10]:

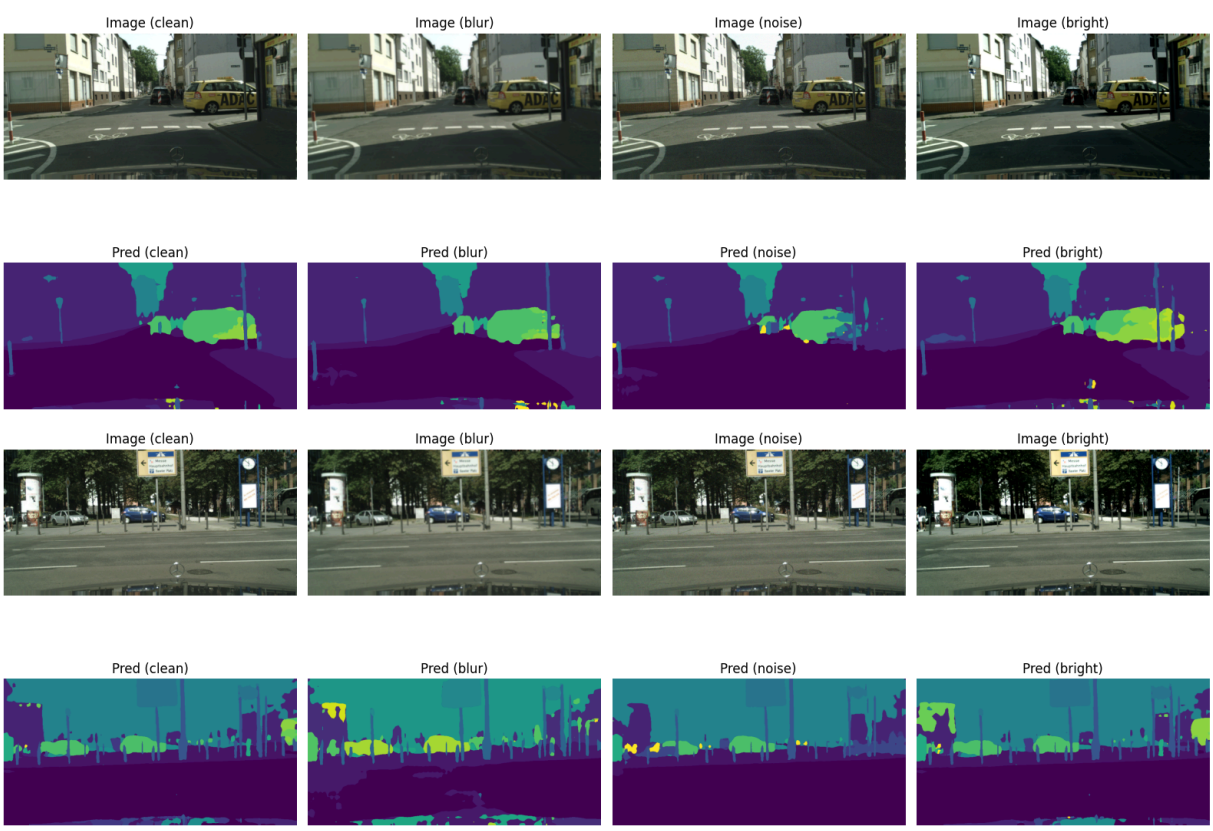
```python
@torch.no_grad()
def predict_one(model, x):
    model.eval()
    x = x.unsqueeze(0).to(device)
    logits = model(x)["out"]
    pred = logits.argmax(dim=1).squeeze(0).cpu()
    return pred

def denorm_img(x):
    # Undo ImageNet normalization for display.
    mean = torch.tensor([0.485, 0.456, 0.406]).view(3, 1, 1)
    std = torch.tensor([0.229, 0.224, 0.225]).view(3, 1, 1)
    img = x.cpu() * std + mean
    return torch.clamp(img, 0, 1)
```

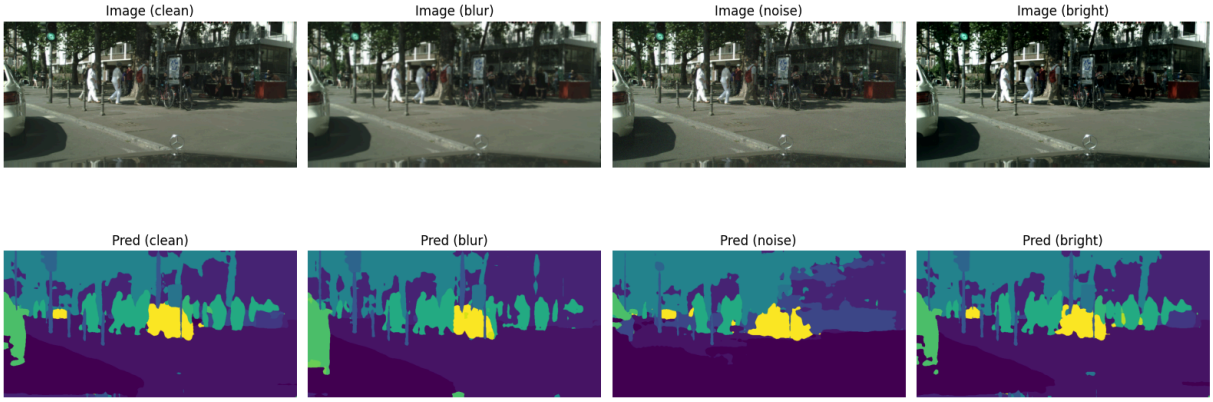
Show side-by-side for a few examples

In [11]:

```python
def show_example(idx=0):
    x_clean, y, _ = val_clean[idx]
```

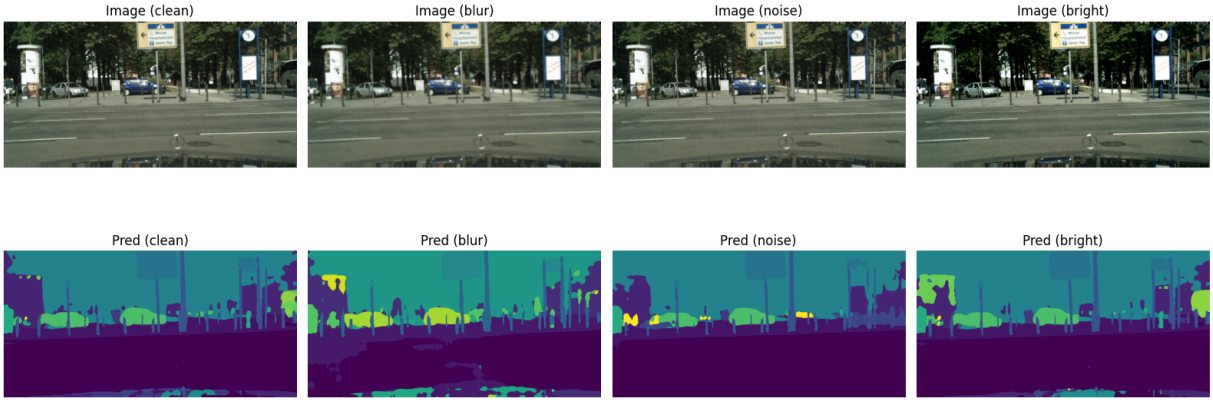
```
    x_blur, _, _ = val_blur[idx]
    x_noise, _, _ = val_noise[idx]
    x_bright, _, _ = val_bright[idx]

    p_clean = predict_one(model, x_clean)
    p_blur = predict_one(model, x_blur)
    p_noise = predict_one(model, x_noise)
    p_bright = predict_one(model, x_bright)

    fig, axes = plt.subplots(2, 4, figsize=(16, 7))

    # Top row: inputs under each condition. Bottom row: model predictions.
    axes[0, 0].imshow(denorm_img(x_clean).permute(1, 2, 0)); axes[0, 0].set_title("
    axes[0, 1].imshow(denorm_img(x_blur).permute(1, 2, 0)); axes[0, 1].set_title("I
    axes[0, 2].imshow(denorm_img(x_noise).permute(1, 2, 0)); axes[0, 2].set_title("
    axes[0, 3].imshow(denorm_img(x_bright).permute(1, 2, 0)); axes[0, 3].set_title(

    axes[1, 0].imshow(p_clean); axes[1, 0].set_title("Pred (clean)")
    axes[1, 1].imshow(p_blur); axes[1, 1].set_title("Pred (blur)")
    axes[1, 2].imshow(p_noise); axes[1, 2].set_title("Pred (noise)")
    axes[1, 3].imshow(p_bright); axes[1, 3].set_title("Pred (bright)")

    for ax in axes.flat:
        ax.axis("off")

    plt.tight_layout()
    plt.show()

show_example(0)
show_example(25)
show_example(100)
```





Save a figure for your report

In [12]:
```
# Save one representative qualitative panel for the report.
idx = 25
x_clean, y, _ = val_clean[idx]
x_blur, _, _ = val_blur[idx]
x_noise, _, _ = val_noise[idx]
x_bright, _, _ = val_bright[idx]

p_clean = predict_one(model, x_clean)
p_blur = predict_one(model, x_blur)
p_noise = predict_one(model, x_noise)
p_bright = predict_one(model, x_bright)

fig, axes = plt.subplots(2, 4, figsize=(16, 7))
axes[0, 0].imshow(denorm_img(x_clean).permute(1, 2, 0)); axes[0, 0].set_title("Imag
axes[0, 1].imshow(denorm_img(x_blur).permute(1, 2, 0)); axes[0, 1].set_title("Image
axes[0, 2].imshow(denorm_img(x_noise).permute(1, 2, 0)); axes[0, 2].set_title("Imag
axes[0, 3].imshow(denorm_img(x_bright).permute(1, 2, 0)); axes[0, 3].set_title("Ima
axes[1, 0].imshow(p_clean); axes[1, 0].set_title("Pred (clean)")
axes[1, 1].imshow(p_blur); axes[1, 1].set_title("Pred (blur)")
axes[1, 2].imshow(p_noise); axes[1, 2].set_title("Pred (noise)")
axes[1, 3].imshow(p_bright); axes[1, 3].set_title("Pred (bright)")
for ax in axes.flat:
    ax.axis("off")
plt.tight_layout()

save_path = OUT / "qual_shift_example.png"
plt.savefig(save_path, dpi=200)
plt.show()
print("Saved:", save_path)
```



Saved: c:\Users\dhruv\temporal_cityscapes\outputs\qual_shift_example.png